



McGill

Department of
Epidemiology, Biostatistics
and Occupational Health

BIOS 640 - Introduction to Health Data Science Methods

Assignment Week 4

Instructions

1. This assignment counts for 18% of your final grade.
2. Late submissions will not be accepted.
3. Your submission must include the following file:
 - A Word or PDF file that contains the link to your GitHub repository.
4. Try to adhere to the naming and styling conventions introduced during Week 1 of the course.
5. Please refer to the assignment rubric to understand how your work will be graded.

Good luck!

Important Notes

About the NHANES Dataset

The NHANES dataset used here is an older one, and it carries a few naming conventions that were once standard but are considered outdated today. Two are worth flagging up front:

- **gender** (originally coded as `RIAGENDR`) actually captures biological sex rather than gender identity. In modern practice, this variable would be labeled `sex`, or would allow for more than two categories.
- **ethnicity** (originally coded as `RIDRETH1` and `RIDRETH3`) describes race, a social grouping based on broadly perceived physical characteristics, rather than ethnicity, another social construct based on shared cultural and/or ancestral background. In modern practice, this variable would more accurately be called `race` or `self-reported race`.

For learning and demonstration purposes, we work with the original NHANES column labels throughout this assignment. When you clean your own datasets in practice, however, you can and should adopt clearer, more current variable naming from the outset.

Building on Assignments 2 and 3

This assignment builds directly on your work from **Assignments 2 and 3**. You will need:

- The **R Markdown report** you built in Assignment 3 (for Exercise 2).
- The **saved plots** from Assignment 3, Exercise 2 (for the dashboard in Exercise 3).
- The **average systolic blood pressure** variable from Assignment 3, Exercise 4 (for the dashboard in Exercise 3).

If any part of these earlier assignments is incomplete, please complete the relevant sections first.

About Reproducibility

Since your submission is a link to your GitHub repository, your work will be graded by cloning the repository. Please verify that:

- All file paths in your R Markdown files use `here()` or relative paths; never absolute paths like `C:/Users/...`.
- Your `README.md` briefly explains how to navigate the repository and reproduce your outputs.
- Your rendered outputs (HTML report, dashboard, PDF report) are committed alongside the source `.Rmd` files, so a grader does not need to render everything from scratch to see your work.

Exercise 1 (25%) Creating a GitHub repository

In this exercise, you will create a public GitHub repository for this week's assignment. It should include all files used in Exercises 2 to 4. Make sure to include the link to your repository in the submission document.

- Create a GitHub repository with a clear and meaningful name.
- Move all relevant files into the repository. These should include your R Markdown files, datasets, figures, and reports.
- Organize the files into appropriately named folders (e.g., data, code, etc.).
- Include a `README.md` file that briefly describes the purpose of the repository and the content of each folder.
- Use meaningful and descriptive commit messages throughout your work.

Exercise 2 (25%) Customizing R Markdown outputs through GitHub branches

In this exercise, you will update your R Markdown report from last week and customize its output.

- Upload the original version of your R Markdown report from last week to your GitHub repository (both the `.Rmd` and `.pdf` files) along with any other files used by the Markdown file (e.g., the dataset).
- Create a new branch to work on your modifications.
- In this new branch, update your R Markdown file (i.e., the `.Rmd` file) to include the following:
 - Change the output format to `html`.
 - Specify a theme and a syntax highlighting style.
 - Include a floating table of contents with appropriate depth so users can jump from one exercise to another.
 - Use tabbed sections in Exercise 1 of the report. One tab should display the individual figures you recreated. A second tab should display the combined figures.
 - At the start of Exercise 2 of the report, include an interactive table using the `DT` package to display the cleaned NHANES dataset and allow users to explore it.
- Render the updated report and upload the `html` file to your new branch.
- Commit your changes and push them.
- Merge your new branch into the main branch.
- After merging, delete your new branch.

Exercise 3 (20%) Creating a dashboard

The goal of this exercise is to create a simple, interactive dashboard using the **flexdashboard** package and the NHANES dataset. Make sure to upload the R Markdown file and the rendered dashboard to your GitHub repository.

Follow the steps below to complete the dashboard.

- Step 1: Create a new R Markdown file and format it as a **flexdashboard** by specifying the output as `flexdashboard::flex_dashboard:` along with the following specifications:
 - Title: “NHANES Dashboard”
 - Orientation: Rows
- Step 2: Add an R chunk to:
 - Load the usual data manipulation packages.
 - Load the **flexdashboard** package.
 - Import the cleaned NHANES dataset.
- Step 3: Define the layout.
 - Create a layout with two rows:
 - * Top row: `{data-height=650}`
 - * Bottom row: `{data-height=350}`
 - Define a single section in the top row and two sections in the bottom row.

Refer to the documentation on [chart layout](#) for help.

- Step 4: In the top row, insert a figure with one of the plots you previously obtained in past assignments that describes the NHANES sample.
 - Make sure that the figure fits the allocated area.
- Step 5: In the bottom row, left column, insert a gauge indicating the percentage of individuals 21 years old or older.
 - Set the range to 0-100 and use color thresholds of your choice for the gauge.
- Step 6: In the bottom row, right column, insert a value indicating the percentage of individuals with an average systolic blood pressure above 120 mm Hg.
 - Use the color of your choice for the value box.

The following functions from **flexdashboard** may be useful:

- `gauge()`
- `valueBox()`

Value boxes and gauges are typically placed inside a chunk beneath a `###` heading; see the **flexdashboard** documentation for the exact syntax.

Exercise 4 (30%) Creating complex tables

In this exercise, you will create a short pdf report using R Markdown to present a description of the cleaned NHANES dataset. Be sure to upload the R Markdown file and the rendered report to your GitHub repository.

Your report should include:

- A brief introduction to the NHANES dataset.
- One or several tables where you applied what you have learned about table design and formatting.
- A bibliography citing the NHANES dataset using the R Markdown citation tools.

The table(s) must cover the following information:

- The overall sample size of each wave.
- The number and percentage of males for each wave.
- The number and percentage of each ethnicity for each wave.
- Summary statistics (minimum, maximum, median, mean, standard deviation) of the average systolic blood pressure, by wave.

Furthermore, your table(s) should:

- Use advanced formatting options to emphasize information of your choice.
- Have a clear title.
- Be clear, informative, and easy to read.

Your report should read as a short narrative. Therefore, it should include written sections describing and summarizing the main information from your table(s).

Hints:

- Use `kable()` with `kableExtra` for formatting. Since the report is PDF, stick to LaTeX-friendly options (`add_header_above()`, `column_spec()`, `row_spec()`, `footnote()`) since HTML features like hover won't render.
- Aggregate first with `dplyr` (`group_by(wave) + summarise(...)`), then pass the summary to `kable()`.